



GRAVITATION

The force that binds the Universe together.

1. INTRODUCTION

- Gravitation is a universal force of attraction between any two objects having mass.
- It acts along the line joining the centres of the two bodies.
- It is the weakest of all fundamental forces, but it has infinite range.

KEY POINTS

- Universal
- Attractive
- Central
- Long range
- Weak but Always Present

2. NEWTON'S LAW OF UNIVERSAL GRAVITATION

Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G \frac{m_1 m_2}{r^2}$$

Where,

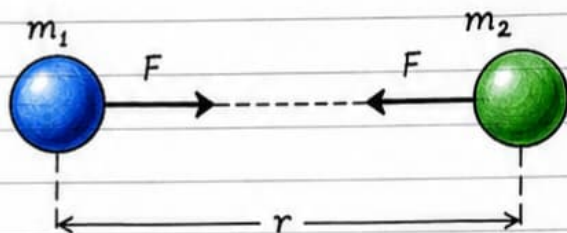
F = Gravitational force (N)

m_1, m_2 = Masses of the bodies (kg)

r = Distance between their centres (m)

G = Universal gravitational constant

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$



Properties of Gravitational Force

- Always attractive.
- Acts along the line joining the centres.
- Obeys superposition principle.
- Much weaker than other forces.

3. CHARACTERISTICS OF GRAVITATIONAL FORCE



Universal

Acts between all objects in the universe.



Central

Acts along the line joining the centres.



Additive

Net force is the vector sum of individual forces.



Infinite Range

Acts over any distance, no matter how large.

4. DIMENSIONAL FORMULA OF G

$$\text{From, } F = G \frac{m_1 m_2}{r^2} \\ [G] = \frac{[F][r]^2}{[m]^2} = \frac{MLT^{-2} \cdot L^2}{M^2} = M^{-1} L^3 T^{-2}$$

SI Unit of G

$$\text{N m}^2 \text{ kg}^{-2}$$

Remember

Mass $\rightarrow$ kg
Distance $\rightarrow$ m
Force $\rightarrow$ N

5. ACCELERATION DUE TO GRAVITY (g)

- It is the acceleration produced in a body when it is freely falling under the gravitational pull of a planet. It depends on the mass and radius of the planet.

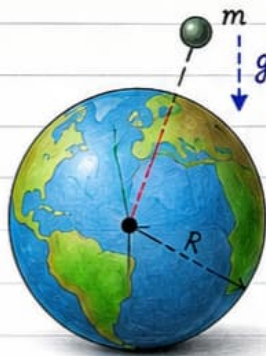
$$g = \frac{GM}{R^2}$$

Where,

M = Mass of the planet (kg)

R = Radius of the planet (m)

G = Universal gravitational constant



Important

The value of g is not same at all places on the Earth.

Value of g on Earth :

At the surface, $g \approx 9.8 \text{ m s}^{-2}$
(mean value)

S.I. Unit : m s^{-2}

Variation of g

- Decreases with altitude.
- Decreases with depth.
- Maximum at poles.
- Minimum at equator.

Reason

- Distance from the centre increases.
- Mass enclosed within decreases.
- Earth is slightly oblate.
- Centrifugal force is maximum.

Remember



- g depends on
- (i) Mass of planet
 - (ii) Radius of planet
 - (iii) Position on / inside the planet

6. GRAVITATIONAL POTENTIAL ENERGY (U)

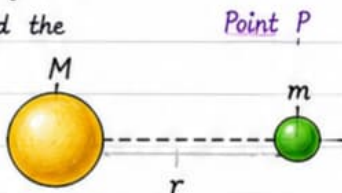
- The work done in bringing a mass m from infinity to a point at a distance r from a mass M is called the gravitational potential energy of the system.

$$U = -\frac{GMm}{r}$$

S.I. Unit : Joule (J)

Sign Convention :

- $U < 0$ (Bound system)
- $U = 0$ (At infinity)
- $U > 0$ (Not possible in gravitation)



Key Points

- U is negative.
- U is zero at infinite distance.
- As r decreases, U becomes more negative.

7. ESCAPE VELOCITY (v_e)

- The **minimum** velocity required by a body to escape from the gravitational field of a planet (without further propulsion) is called **escape velocity**.

$$v_e = \sqrt{\frac{2GM}{R}}$$

Where,

M = Mass of the planet (kg)

R = Radius of the planet (m)

G = Universal gravitational constant



For Earth,

$v_e \approx 11.2 \text{ km s}^{-1}$

8. ORBITAL VELOCITY (v_o)

- The **minimum horizontal velocity** needed by a body so that it revolves around a planet in a **circular orbit** close to its surface.

$$v_o = \sqrt{\frac{GM}{R}}$$

For Earth,

$v_o \approx 7.9 \text{ km s}^{-1}$



★ QUICK SUMMARY ★

- Gravitation $\rightarrow$ Universal attractive force between masses.
- $F = G \frac{m_1 m_2}{r^2} \rightarrow$ Inverse square law.
- $g = \frac{GM}{R^2} \rightarrow$ Acceleration due to gravity.
- $U = -\frac{GMm}{r} \rightarrow$ Potential energy.
- $v_e = \sqrt{\frac{2GM}{R}} \rightarrow$ Escape velocity.
- $v_o = \sqrt{\frac{GM}{R}} \rightarrow$ Orbital velocity.

★ **FUN FACT**

A satellite in orbit is continuously falling towards Earth but never reaches it! 😊

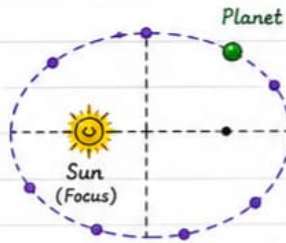


9. KEPLER'S LAWS OF PLANETARY MOTION

Johannes Kepler (1609-1619) stated three empirical laws based on careful study of planetary motion.

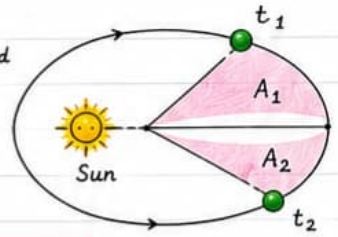
(i) LAW OF ORBITS

Each planet moves in an **elliptical orbit** with the Sun at one of the foci.



(ii) LAW OF AREAS

The line joining a planet and the Sun sweeps out equal areas in equal intervals of time.



$$\text{If } t_1 = t_2, \text{ then } A_1 = A_2$$

(iii) LAW OF PERIODS

The square of the time period (T) of revolution of a planet is directly proportional to the cube of the semi-major axis (a) of its orbit.

$$T^2 \propto a^3 \rightarrow \frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3}$$

Important

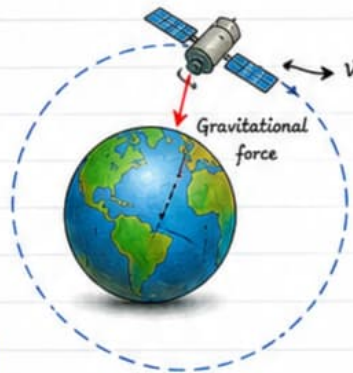
Kepler's laws are valid for all planets in the solar system.

10. SATELLITES AND APPLICATIONS OF GRAVITATION

(i) Artificial Satellites

- A satellite is a body which revolves around a planet in a **circular orbit** due to gravitational pull.
- Gravity provides the centripetal force.
- For a satellite of mass m moving in a circular orbit of radius r around a planet of mass M ,

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \rightarrow v = \sqrt{\frac{GM}{r}}$$



(ii) Applications of Gravitation

- ✓ Motion of planets and comets.
- ✓ Orbits of artificial satellites.
- ✓ Ocean tides due to Moon and Sun.
- ✓ Weight of objects on the Earth.
- ✓ Escape of gases from a planet's surface.

11. GRAVITATION NEAR THE EARTH'S SURFACE

- Consider a body of mass m at a height h above the Earth's surface.
- Distance from the centre of Earth, $r = R + h$
- Acceleration due to gravity at height h ,

$$g_h = \frac{GM}{(R+h)^2}$$



Special Cases

- At the surface ($h = 0$)

$$g = \frac{GM}{R^2} = 9.8 \text{ ms}^{-2}$$

- At height h ($h \ll R$)

$$g_h \approx g \left(1 - \frac{2h}{R}\right)$$

- At depth d below surface

$$g_d = g \left(1 - \frac{d}{R}\right)$$

Remember

- Value of g is maximum at the poles.
- Value of g is minimum at the equator.
- g decreases with height.
- Inside the Earth, g decreases linearly with depth and becomes zero at the centre.

KEY FORMULAE

- $F = G \frac{m_1 m_2}{r^2}$
- $g = \frac{GM}{R^2}$
- $U = -\frac{GMm}{r}$
- $v_e = \sqrt{\frac{2GM}{R}}$
- $v_o = \sqrt{\frac{GM}{R}}$

UNITS

- $G \rightarrow \text{Nm}^2 \text{kg}^{-2}$
- $g \rightarrow \text{ms}^{-2}$
- $U \rightarrow \text{J}$
- Velocity $\rightarrow \text{ms}^{-1}$

DIMENSIONS

- $[G] = \text{M}^{-1} \text{L}^3 \text{T}^{-2}$
- $[g] = \text{LT}^{-2}$
- $[U] = \text{ML}^2 \text{T}^{-2}$

THINK !

Why does a satellite remain in orbit and not fall on Earth?
→ Because its tangential velocity is just right to keep falling around Earth.

FUN FACT

The Moon stays in orbit around Earth because it is constantly "falling" towards Earth but never reaches it!

12. WEIGHT OF A BODY

- The gravitational force exerted by the Earth on a body is called its **weight**.
- It acts towards the **centre** of the Earth.

Formula

$$W = mg$$

Where,

m = Mass of the body (kg)

g = Acceleration due to gravity
(9.8 m s^{-2} on Earth)

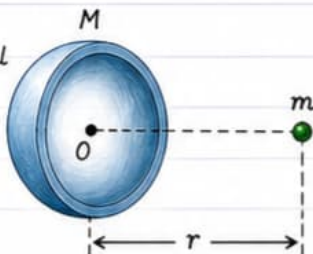
Think!

An object has mass everywhere in the universe, but its weight changes from place to place.

13. GRAVITATION BETWEEN SPHERICAL BODIES (SHELL THEOREM)

- (i) **Outside a spherical shell :**

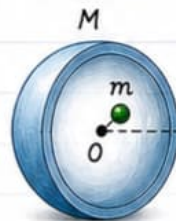
A spherical shell attracts an external mass as if its entire mass were concentrated at its centre.



$$F = G \frac{Mm}{r^2}$$

- (ii) **Inside a spherical shell :**

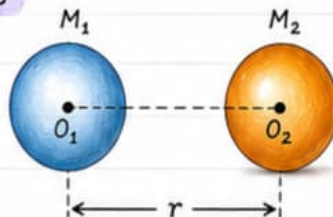
The gravitational force inside a spherical shell is zero everywhere.



$$F = 0$$

14. GRAVITATION BETWEEN TWO SPHERICAL SHELLS

The gravitational force between two spherical shells depends only on their total masses and the distance between their centres, as if the masses were concentrated at their centres.



$$F = G \frac{M_1 M_2}{r^2}$$

15. NUMERICAL EXAMPLES

- ① Two masses 3 kg and 5 kg are placed 2 m apart in vacuum. Find the gravitational force between them.

Given : $m_1 = 3 \text{ kg}$, $m_2 = 5 \text{ kg}$, $r = 2 \text{ m}$

$$F = G \frac{m_1 m_2}{r^2} = 6.67 \times 10^{-11} \frac{3 \times 5}{(2)^2}$$

$$F = 2.50 \times 10^{-10} \text{ N}$$

- ② Calculate the acceleration due to gravity on a planet whose mass is 8 times that of Earth and radius is 2 times that of Earth.

$$g_p = \frac{GM_p}{R_p^2} = \frac{G(8M_E)}{(2R_E)^2} = \frac{8}{4} \frac{GM_E}{R_E^2} = 2g_E$$

$$\text{So, } g_p = 2g_E = 19.6 \text{ m s}^{-2}$$

Note ★

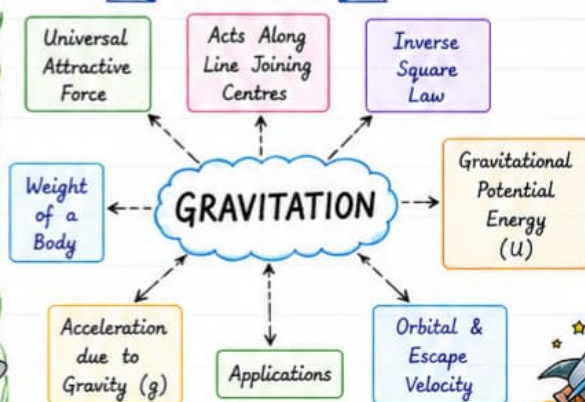
Gravity is a very weak force, but it acts over very large distances.

QUICK REVISION

- Gravitation is a universal attractive force between any two masses.
- It acts along the line joining their centres.
- $F = G \frac{m_1 m_2}{r^2}$ (Inverse square law)
- $g = \frac{GM}{R^2}$ (Acceleration due to gravity)
- $U = -\frac{GMm}{r}$ (Potential energy)
- $v_e = \sqrt{\frac{2GM}{R}}$ (Escape velocity)
- $v_o = \sqrt{\frac{GM}{R}}$ (Orbital velocity)
- Weight, $W = mg$
- Inside a shell $\rightarrow F = 0$; Outside a shell $\rightarrow$ Acts as if mass at centre.



CONCEPT MAP



★ Gravitation holds the universe together → from falling apples to revolving galaxies. ♥

THE END 😊